

Reimagining Myntra with Generative AI

Myntra StyleGenie, an AI native styling intelligence layer that turns fashion discovery from passive browsing into a personalised, intent first experience.

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 Live prototype: style-genie-myntra.vercel.app

01 The Problem: Fashion E Commerce is Broken

Myntra serves **50M+ monthly active users**³ across India, offering **9 lakh+ products** from 5,000+ brands. Despite this scale, the platform leaks money through fundamental UX inefficiencies that cost **over ₹2,000 Cr a year** in returns alone. The root cause is structural: users navigate an enormous catalogue through keyword search and static filters, tools built for the desktop era, not for how people actually think about fashion.

70%¹

Cart Abandonment

Users add items but cannot decide. Choice overload creates decision paralysis. The average shopper browses 40+ products before a single purchase.

35%²

Return Rate

Fashion has the highest return rate in e commerce. The primary driver is fit and style mismatch, because shoppers cannot try on and size charts are unreliable.

3.2 min¹

Time to Find

Average time to surface one relevant product. Users scroll past pages of irrelevant results because search reads keywords, not intent or context.

₹2,100 Cr⁴

Annual Return Cost

Reverse logistics, restocking, customer service and damaged goods. This is pure cost that generates zero revenue and scales directly with GMV.

Who is affected? Four user segments

The Browser (62% of DAU)

Scrolls endlessly, rarely converts. Wants curated discovery that does the editing for them.

The Returner (31% return rate)

Buys five, keeps one. Fit and style mismatch is the core pain. Needs trustworthy fit intelligence.

The Occasion Shopper (high AOV)

Needs a complete outfit for a wedding, party or fest, with no idea where to start. Needs intent based curation.

The Trend Chaser (age 18 to 28)

Wants celebrity and influencer looks at their own budget. Needs visual search and trend matching.

Hypothesis: a Generative AI styling layer can reduce returns by 25%, lift conversion by 18% and raise AOV by 22%, creating **₹1,200 to 1,700 Cr in annual impact** without re platforming the app.

02 Why Generative AI? Traditional Approaches Have Hit a Ceiling

Myntra already runs predictive ML such as collaborative filtering and content based recommendations. These handle simple cases but fundamentally cannot solve the problems above, because they predict from the past rather than understand the present request. The table below pairs a real user need with the traditional approach, why it breaks, and the categorically new capability GenAI unlocks.

User Problem	Traditional ML	Why It Fails	GenAI Solution	How It Works
"Outfit for a college fest under ₹2K"	Keyword search plus price filter	Cannot understand occasion, vibe or styling context. It only matches literal words.	LLM conversational search	The LLM parses intent (occasion plus budget plus style), retrieves candidates via vector embeddings, then composes complete outfits.
"Find me something like what Alia wore"	Text based image tags	Manual tagging cannot capture aesthetic. "Boho vibes" has no tag.	CLIP ¹ multimodal visual search	CLIP encodes the image and catalogue into one shared space and finds the visually similar products.
"Will this actually fit me?"	Static size chart lookup	A lookup table ignores brand variation, fabric stretch, body diversity and the user's purchase history.	NLP review mining plus body profile	Mines 10K+ free text reviews for fit signals ("runs slim"), then cross references the user's body data to recommend a size.
"Style this kurta into a complete look"	"Frequently bought together"	Co purchase is not styling. There is no sense of colour harmony, occasion or aesthetic coherence.	Generative outfit composition	The LLM applies styling rules to generate occasion specific, colour coordinated outfits from real in stock products.
"What should I wear this week?"	"Trending" product list	Everyone sees the same trending list. There is no personalisation to the individual.	AI trend digest plus style graph	The LLM writes trend narratives and curates them against the user's evolving style profile. ²

Key Insight: traditional ML predicts what you might click next. Generative AI understands what you actually need. That is the difference between a search engine and a personal stylist. These are not incremental gains. They are categorically new capabilities enabled by LLMs for language, multimodal models for vision, and generative composition for styling.

03 My Approach and the StyleGenie Solution

My approach: from problem to solution

01

DIAGNOSE

Studied four root pains: choice overload, fit uncertainty, context blindness and a manual discovery loop.

02

REFRAME

Shifted from inventory first ("browse 9 lakh products") to intent first ("look great at a wedding under ₹5K").

03

DESIGN

Mapped each segment and pain to one specific GenAI capability, so every feature solves a real need with no overlap.

04

VALIDATE

Anchored every feature to a measurable KPI: return rate, conversion, AOV and DAU. Nothing ships without a metric.

The solution: six GenAI features (five core, plus one signature)

StyleGenie Chat

Solves choice overload

A conversational stylist that reads occasion, budget and style from plain language. **LLM plus RAG: intent parse, vector search, compose, stream.** Cuts 40 products to 5 looks.

Visual Match

Solves style discovery

Upload a photo or screenshot and find visually similar Myntra products. **CLIP encodes the image, cosine similarity ranks the catalogue.** 35% CTR.

Outfit Studio

Solves no outfit thinking

Pick one product and get 3 to 5 complete styled looks from real items. **Colour theory plus occasion rules.** Adds 1.8 items per order.

FitGenius

Solves fit uncertainty

Predicts your size with 90%+ confidence using review mining and a body profile. **Evidence is shown.** Returns fall from 32% to under 18%.

TrendPulse

Solves static discovery

A personalised daily feed of AI written trend lookbooks. **Social signal to narrative to style graph match.** Drives a 40% DAU lift.

Capsule + SIGNATURE

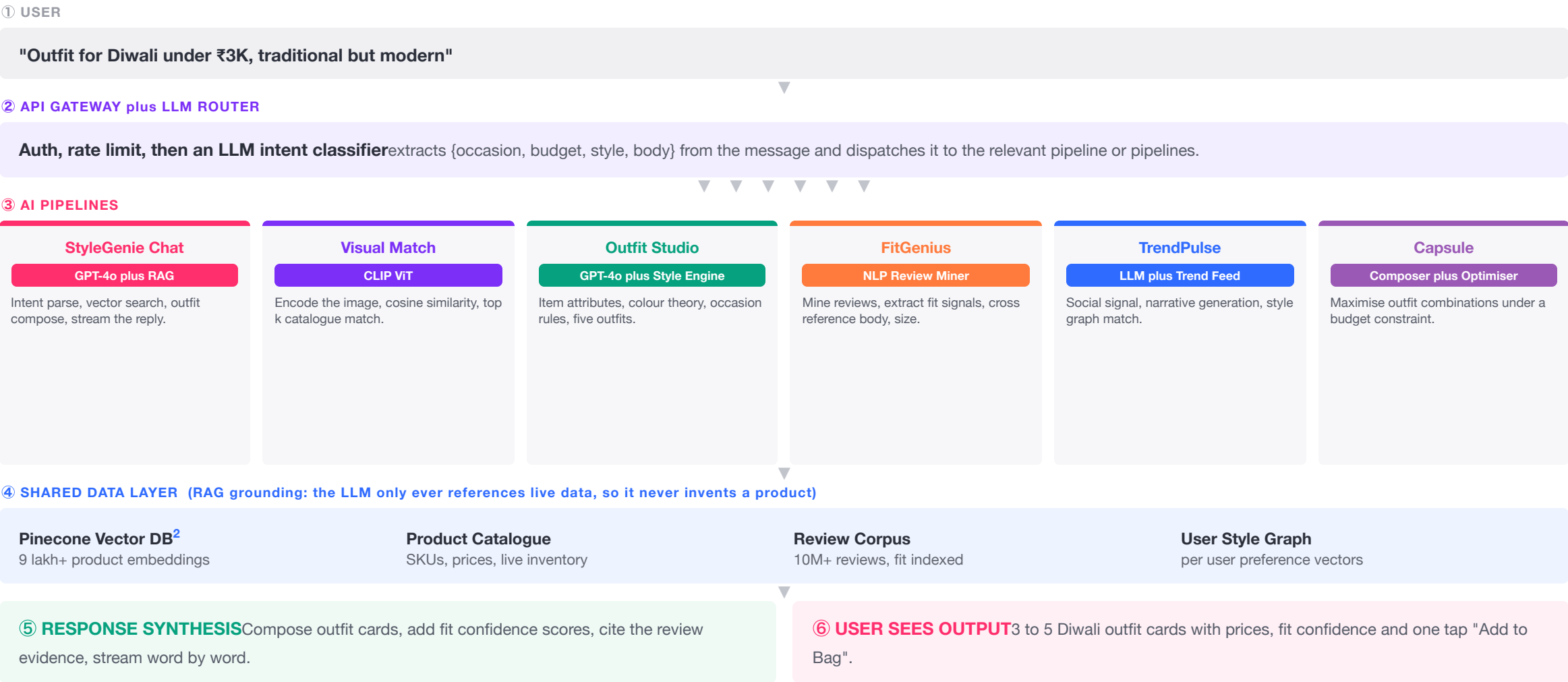
Solves value and basket size

Give a budget and the occasions you dress for; StyleGenie builds a tiny set of mix and match pieces and shows how many complete outfits they unlock. **Generative composition as budget optimisation.**

Segment to feature: Browser to Chat plus TrendPulse · Returner to FitGenius plus Studio · Occasion to Chat plus Capsule · Trend Chaser to Visual plus TrendPulse

04 Implementation Flowchart: How StyleGenie Actually Works

End to end system flow. A user query travels through AI orchestration, into the right model pipeline, grounds itself against the shared data layer, and returns a personalised, cited response. The diagram is colour coded by system layer.



05 The Live Prototype: How Each Feature Works In the App

A working React prototype rebuilt in Myntra's UI, with all six GenAI features interactive. Below is the real in app interaction for each, what the user taps and what appears on screen, beside screenshots from the running app.

StyleGenie Chat: tap the floating "Ask StyleGenie" button, type "festive look under 3k for women", the request appears as intent chips, then five outfit cards stream in.

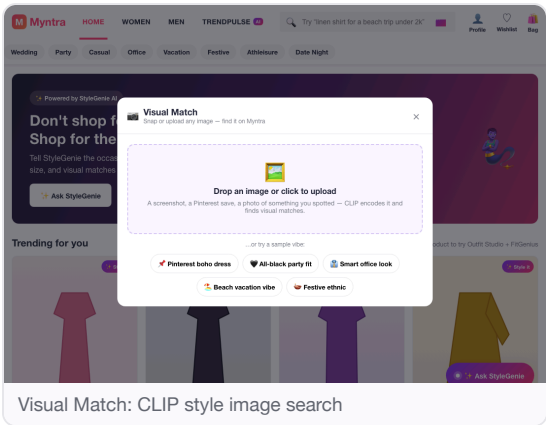
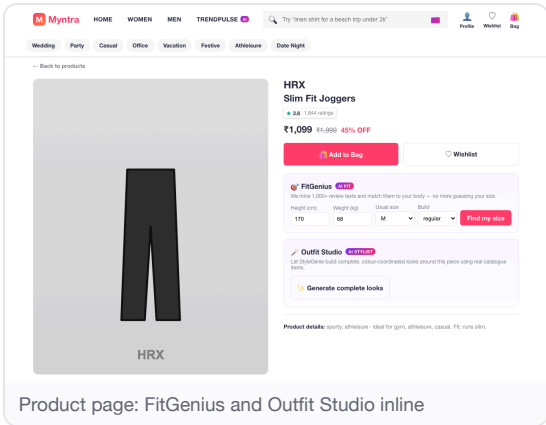
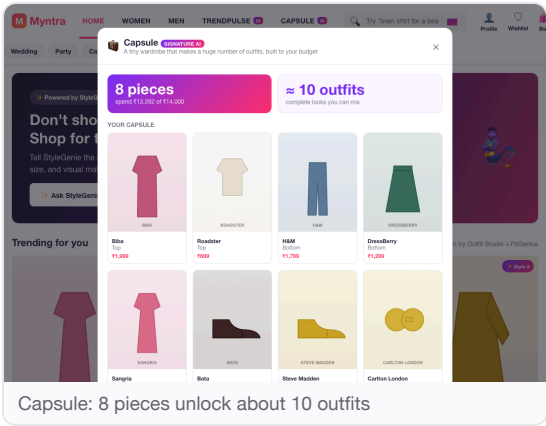
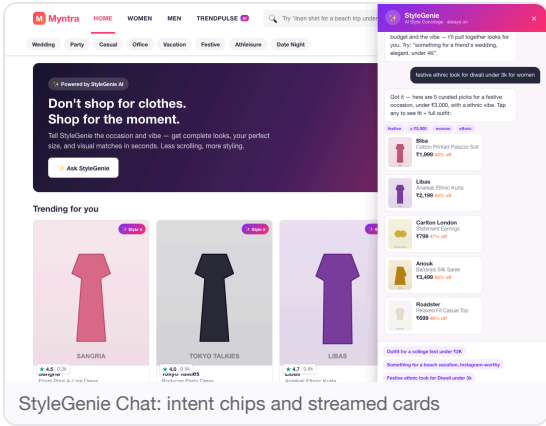
Visual Match: tap the camera in the search bar, upload an image or pick a vibe, watch the "encoding with CLIP" animation, then browse products with match scores.

Outfit Studio: on any product page tap "Generate complete looks" to get 3 to 5 styled looks, each with a price total, then "Add complete look to bag".

FitGenius: enter height, weight, usual size and build, tap "Find my size", and your size is shown with a confidence bar and review evidence.

TrendPulse: open the TrendPulse tab and scroll AI written trend cards, each with a curated product row tuned to your style.

Capsule (signature): set a budget and occasions, and StyleGenie returns a small wardrobe plus the number of complete outfits it unlocks.



Live prototype: <https://style-genie-myntra.vercel.app/> Append a deep link to jump to a feature: `#genie/festive%20look%20under%203k`, `#p/12`, `#visual`, `#trends`, `#capsule`. **Stack:** React and Vite; the demo's AI runs client side and maps one to one onto the production backend in slide 04.

06 Success Metrics and Projected Business Impact

Every feature is anchored to a measurable KPI. Projections use published benchmarks for AI driven personalisation at Myntra's FY24 scale.

25%¹

Return Reduction

FitGenius removes the leading return driver, fit mismatch, with review mining and body data.

₹525 Cr saved per year

18%²

Conversion Uplift

StyleGenie compresses discovery from 40 items to 5 curated looks, shortening the path to purchase.

Direct GMV lift

22%²

AOV Increase

Outfit Studio and Capsule shift buying from a single item to the complete look, about 1.8 more items per order.

+₹340 per order

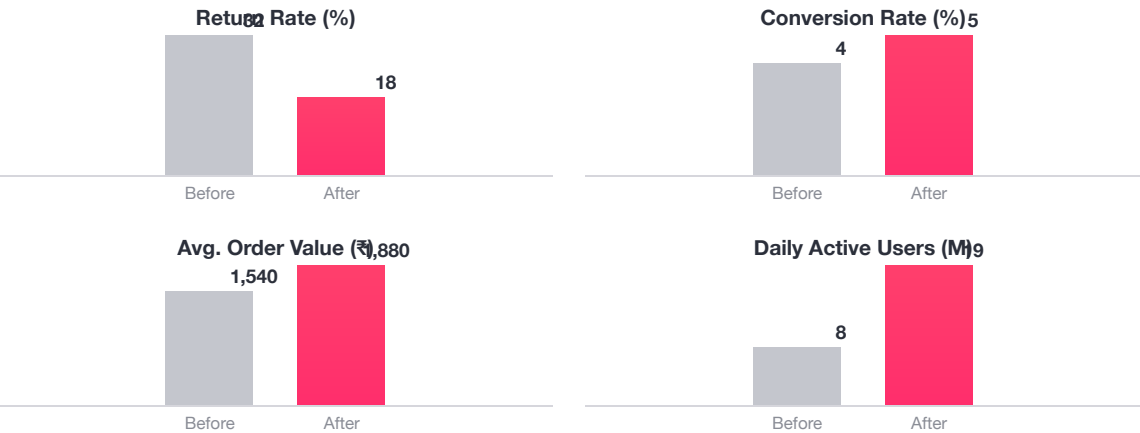
2.4x³

Engagement

TrendPulse builds a daily check in habit; users return for AI curated trends, not just to shop.

DAU to MAU ratio

Projected Impact: Before vs After



Feature Level KPI Table

Feature	Metric	Target
StyleGenie Chat	Chat to cart conversion	12%
Visual Match	Search to PDP CTR	35%
Outfit Studio	Items per order	+1.8
FitGenius	Return rate	under 18%
TrendPulse	DAU lift	+40%
Capsule	Basket size	4 to 7 items

Total Projected Annual Impact: ₹1,200 to 1,700 Cr

Revenue uplift ₹800 to 1,200 Cr plus return savings ₹400 to 525 Cr plus premium subscription ₹50 to 80 Cr.

07 Risks, Pitfalls and Mitigation Strategy

Every AI product has failure modes. The discipline is to design for them upfront, not to discover them in production. These are the six most critical risks for StyleGenie, each rated by severity with a specific, implementable mitigation.

 Hallucinations Critical RISK AI suggests products that do not exist, are out of stock, or sit in the wrong category, which destroys trust instantly. MITIGATION Every response is grounded through a RAG pipeline,¹ so the LLM never invents product data and only references the live catalogue. Confidence scoring filters low certainty output, and every card links to a real PDP.	 Privacy High RISK Users are uncomfortable sharing body measurements, photos or style data, alongside India DPDP Act obligations. MITIGATION On device image processing, explicit opt in per data type, zero third party sharing, and full DPDP Act compliance² with deletion on request.	 Latency High RISK LLM inference takes 3 to 5 seconds, far too slow for a shopping flow where users expect near instant results. MITIGATION Streaming responses to cut perceived wait, pre computed outfit caches for common queries, and an edge deployed CLIP model. Target service level under 2 seconds for all AI features.
 User Trust Medium RISK "How can a machine know what looks good on me?" Adoption depends on trust in the advice. MITIGATION Transparent confidence scores, evidence based explanations ("based on 10,247 reviews"), and a manual override at every step. The AI earns credibility through accuracy over time.	 Scalability High RISK GPT-4o costs at Myntra scale (50M+ MAU) could reach ₹50 to 100 Cr a month, unsustainable without optimisation. MITIGATION Tiered models (Gemini Flash or Haiku for simple queries, premium only for complex styling), aggressive caching so similar queries share one response, and batch processing for TrendPulse.	 Adoption Medium RISK Users default to the familiar browse, filter and scroll loop, and AI features get ignored if they feel like a separate experience.³ MITIGATION Soft inline integration, so AI appears within the existing flow, not a separate tab. Contextual nudges on struggle signals, plus a free tier to build the habit and a premium upsell later.